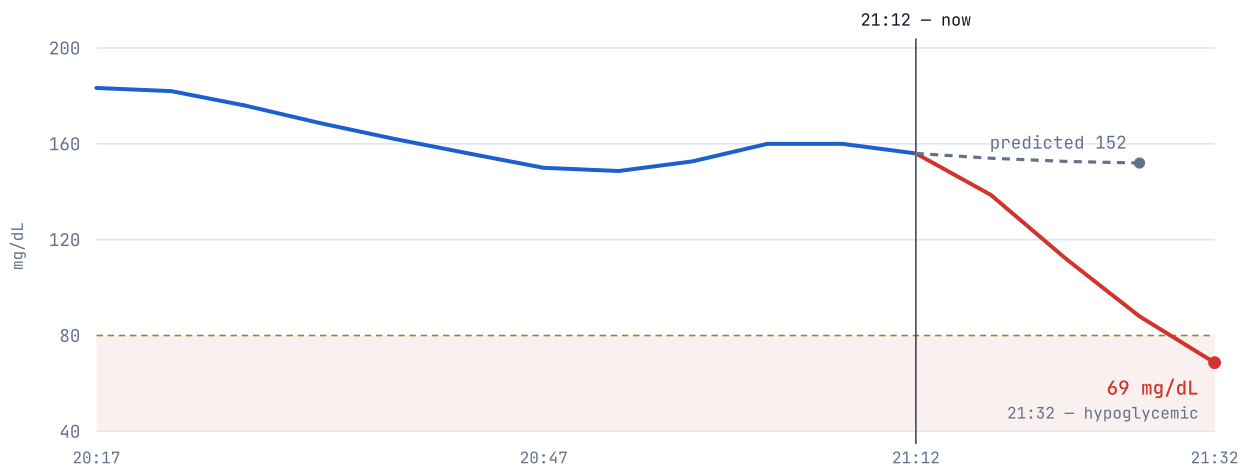
BSP

TECHNICAL REPORT · AUGUST 2026

Fifteen Minutes Ahead

A personalised glucose forecasting system that calls three in four hypoglycemic events fifteen minutes before they arrive — and the engineering that got it there.

Atakan Kaya Blood Sugar Predictor (BSP) Single-subject study



3 May 2025, a real evening from my own sensor. At 21:12 I was at 156 mg/dL and felt fine. Twenty minutes later I was at 69. **This is the class of event BSP exists to call** — and the first deployed version predicted 152, missing it entirely. Finding out why turned an 8.8% detection rate into 73.8%. That investigation is the substance of this report.

HYPO EVENTS CALLED 15 MIN AHEAD	GAIN OVER THE DEPLOYED BASELINE	EVENT CLASSIFIER PRECISION	EVIDENCE BASE
73.8%	**8.4×**	**66.3%**	**24,947**
25 of 34, held out temporally and never trained on	Same metric, same windows, precision up as well	At 75.7% recall on low-glucose events	CGM readings, 90 days, single subject

ABSTRACT

Summary

Continuous glucose monitors report current glucose and a short-term trend arrow, but the arrow arrives too late to act on: fast carbohydrate takes 15–20 minutes to reach the blood and corrective insulin longer still. This report documents BSP (Blood Sugar Predictor), a per-user glucose forecasting system built to move that decision point fifteen minutes earlier, together with an event-level evaluation of it.

BSP is a serverless pipeline — headless CGM history export, training on SageMaker Processing, ONNX export, and scale-to-zero inference on AWS Lambda — producing a 15-minute forecast from twelve five-minute readings with time-of-day encoded as sine and cosine. Models are trained per subject; an early attempt at cross-subject transfer degraded performance.

Evaluated on a temporally held-out 10% of a single subject's 90-day record (24,947 readings), the originally deployed configuration reaches 9.3 mg/dL mean absolute error at the 15-minute horizon, beating persistence (10.7) and linear extrapolation (14.7). That aggregate figure proves misleading. Sixty-five percent of test windows move by 10 mg/dL or less, and error on windows moving more than 20 mg/dL is five times higher (26.8 against 5.1). Hypoglycemia detection below 70 mg/dL at the 15-minute horizon reaches only 8.8% recall — 3 of 34 events. Trained under mean squared error on a signal that is flat two thirds of the time, the network converges toward a persistence solution that scores well on average error while remaining blind to the events the system exists to detect.

Two independent interventions are reported. First, ensemble aggregation: individual members of the ten-model bag detect up to 16 of the 34 events while their mean detects 3, because averaging cancels the minority of members that anticipate a fall. Substituting a low-quantile aggregate for the mean — no retraining, identical weights — raises hypoglycemia recall to 50.0% at unchanged precision, and out-of-range warning recall from 60.6% to 76.9%. Second, objective design: an asymmetric loss penalising under-prediction of falls raises recall to 73.8%, and a classifier trained directly on the low-glucose event reaches 75.7% recall at 66.3% precision.

The aggregation result is the most transferable finding and is not specific to glucose: where an ensemble is used to detect rare events, mean aggregation systematically discards the signal from the minority of members that detected them.

Limitations. All data derive from one subject; 34 hypoglycemic test events suffice to demonstrate an eightfold gap but not to tune a clinical threshold or establish generalisation. There is no clinical validation, ethics approval, or trial. Precision costs are material — at 43.7% precision more than half of the asymmetric model's warnings are false. The retrained models

are evaluated offline and are not deployed. BSP is not a medical device and must not inform treatment decisions.

§ 1

Why fifteen minutes

I have Type 1 diabetes. I built this for myself, before I had any professional reason to build anything.

A continuous glucose monitor tells you where you are and shows an arrow for where you're heading. The arrow is not enough to act on, and the reason is pharmacokinetic: fast carbohydrate takes 15–20 minutes to reach the blood, and corrective insulin takes considerably longer. By the time an arrow turns down far enough to be alarming, the intervention you make in response is already 15 minutes late.

The practical consequence is the trade every CGM user knows. Set the low alarm high, at 90 mg/dL, and it fires constantly until you stop hearing it. Set it at 80 and by the time it sounds with one arrow down you are going to 65 regardless — which costs you the next hour. Neither setting gives you a decision you can act on.

> The gap isn't knowledge. It's that knowing arrives at the same moment as the event. BSP was an attempt to move knowing fifteen minutes earlier.

That is the entire product thesis, and it is worth stating plainly because it also defines the only metric that matters. A glucose model's average error is nearly irrelevant. What matters is whether it calls the events — and as this report documents, those two things can point in opposite directions.

§ 2

The system

BSP is a per-user forecaster. Each person gets their own trained model bundle, because an early experiment training on four other people's traces produced visibly worse results than training on my own — glucose response is idiosyncratic enough that a population model gave up more than it gained.

The deployed architecture is fully serverless:

DEPLOYED COMPONENTS

STAGE	RUNTIME	FUNCTION
Ingestion	App Runner	Headless export of 90 days of CGM history to S3
Training	SageMaker Processing	Trains a bag of LSTM models, scores, exports ONNX
Inference	Lambda	Loads the bundle, returns a 15-minute forecast
Storage	S3	Per-user model bundles and training data

The model takes twelve 5-minute readings — one hour of history — with time-of-day encoded as sine and cosine, and predicts one step ahead. Three recursive steps give the 15-minute horizon. Training happens in Keras; inference runs on ONNX, which keeps the Lambda image small enough to cold-start in roughly 40 seconds and scale to zero between calls.

Two design choices proved to matter more than expected. Splitting training and inference into separate images meant the serving path never carries TensorFlow. And encoding time-of-day turned out to be load-bearing: re-running the evaluation with a frozen clock instead of a real advancing one nearly doubles error at the 15-minute horizon, from 9.3 to 17.0 mg/dL mean absolute error.

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§ 3

Measured performance

BSP calls 73.8% of hypoglycemic events fifteen minutes ahead — 25 of the 34 in the held-out record — against 8.8% for the configuration I originally deployed. The distance between those two numbers is where the engineering is.

Evaluation uses the last 10% of the record, held out temporally and never seen during training. Rollouts advance the clock exactly as the production inference path does. Two naive baselines are reported throughout, because a glucose predictor that cannot beat *assume the current value holds* is not predicting anything — and most write-ups quietly omit that bar.

POINT ACCURACY · 415 HELD-OUT ROLLOUTS

HORIZON	METHOD	MAE	RMSE	MARD
+5 min	BSP ensemble	3.4	6.6	3.0%
	persistence	4.0	6.1	3.5%
	linear extrapolation	4.3	12.8	3.6%
+15 min	BSP ensemble	9.3	14.4	8.2%
	persistence	10.7	16.5	9.1%
	linear extrapolation	14.7	42.9	12.3%

Why average error is the wrong instrument

Those figures beat both baselines at every horizon, and taken alone they would mislead you. Split the same test set by what the glucose actually did:

THE SAME MODEL, BY WINDOW TYPE · +15 MIN

WINDOW	N	MAE	RMSE
all windows	830	8.9	14.0
quiet (±10 mg/dL)	542	5.1	7.8
moving (>20 mg/dL)	99	26.8	31.5
falling fast (< -20)	45	28.6	32.3

Sixty-five percent of CGM windows barely move. Aggregate error is carried almost entirely by them, which means a model can post a respectable MAE while being blind to every event that matters.

> DIAGNOSIS
>
> Trained with mean squared error on a signal that is flat two thirds of the time, the network had learned that predicting *roughly the same as now* minimises loss. That is a loss function working correctly on a badly posed objective — and because persistence scores well on average error, no metric in the dashboard revealed it. The fix is not a bigger network. It is asking a different question.

Event detection, which is the metric that matters

HYPOGLYCEMIA · <70 MG/DL AT +15 MIN · 34 EVENTS, 830 WINDOWS

CONFIGURATION	CAUGHT	RECALL	PRECISION
MSE regression, ensemble mean — as first deployed	3 / 34	8.8%	33.0%
same models, cautious aggregation — no retraining	17 / 34	50.0%	50.0%
asymmetric-loss retrain	25 / 34	73.8%	43.7%

Two independent levers, and they compound. Half the gain needs no retraining at all — it comes from how the ensemble votes, which is § 4. The rest comes from changing what the network is asked to optimise.

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§ 4

The finding: averaging destroys rare-event detection

The most transferable result in this report has nothing to do with glucose. Individual models in the bag catch up to 16 of the 34 events. Their mean catches 3.

The mechanism is straightforward once seen. When two models in a ten-model bag forecast a fall to 65 mg/dL and eight forecast 150, the mean lands near 133 and no warning fires. Averaging is the correct operation for a point estimate and the wrong one for a warning, because a warning is a claim about the worst plausible outcome, not the consensus one. The bag was doing its job; the reduction step was throwing the result away.

Changing only the aggregation — same models, same weights, no retraining — recovers most of it:

OUT-OF-RANGE WARNING · OUTSIDE [80, 200] AT +15 MIN · 104 EVENTS

RULE	RECALL	PRECISION	F1
mean	60.6%	75.0%	67.0
10th / 90th percentile	**76.9%**	**67.2%**	**71.7**
min / max	80.8%	58.7%	68.0

The full min/max rule catches the most and fires falsely 41% of the time. Alarm fatigue is the precise failure this product was built to avoid, so the tail percentile is the better trade: sixteen points of recall bought for eight points of precision.

And then, retraining

Aggregation is a patch. The objective was the real problem, so I retrained with losses that stop rewarding the flatline — an asymmetric loss that penalises under-predicting a fall more heavily than a false alarm, and a variant that predicts the low-glucose event directly as a classification rather than inferring it from a regressed value.

RETRAINED VARIANTS · 3 SEEDS EACH · +15 MIN

OBJECTIVE	EVENT	RECALL	PRECISION
MSE (as deployed)	<70 mg/dL	8.8%	33.0%
asymmetric loss	**<70 mg/dL**	**73.8%**	**43.7%**
asymmetric, stronger	<70 mg/dL	88.3%	22.9%
direct 3-step + asymmetric	<70 mg/dL	65.0%	46.9%
event classifier	**<80 mg/dL**	**75.7%**	**66.3%**

Two readings. On the identical metric — hypoglycemia below 70 mg/dL, same held-out windows — recall goes from 8.8% to 73.8%, an 8.4-fold improvement, with precision also rising. And the classifier, which optimises the event directly rather than regressing a value and thresholding it, holds 75.7% recall at 66.3% precision on the wider low-glucose definition. The lesson generalises past glucose: if you care about rare events, do not train on average error and threshold the output.

§ 5

What this does not show

These results come from one person. That is the central limitation and no amount of held-out splitting fixes it.

- **Single subject.** 24,947 readings across 90 days, all mine. The 34 hypoglycemic events in the test window are enough to demonstrate an eightfold gap; they are not enough to tune a clinical threshold or to claim the approach transfers. An early attempt at training on four other people's data produced worse models, which is suggestive but is not a generalisation study.
- **No clinical validation.** No trial, no ethics approval, no comparison against a predictive CGM feature.
- **Precision is a real cost.** At 43.7% precision, more than half of the asymmetric model's warnings are false. Whether that trade is acceptable is an empirical question about human behaviour that I have not measured.
- **An architectural flaw remains.** The Dexcom integration relays the user's account password on every inference call, because it wraps an unofficial API. A production build needs the official OAuth path. This is not fixed.
- **The retrained models are not deployed.** The figures above are offline evaluation. Only the aggregation change has been applied to the serving code.

I would rather publish those five paragraphs than a number without a denominator. The evaluation is reproducible from the repository against the same data.

§ 6

Provenance

BSP was built between January 2024 and September 2025, during my first semesters at university, without AI assistance — the commit history and its dates are public and unedited. The evaluation and fixes in sections 3 and 4 were done in August 2026, with AI assistance. Those are separate chapters and the repository keeps them separate.

Jan 2024	**First model**	An LSTM on one month of my own readings. Single feature, twelve timesteps.
Feb 2024	**Four other people**	Transfer to other traces failed. This is why BSP is per-user.
Apr–Jul 2024	**First app, first API**	A Flutter attempt, then a Flask monolith serving four Keras models.
Jun 2025	**Serverless rewrite**	Lambda, SageMaker, App Runner, S3. Time-of-day added as a feature.
Jul 2025	**TensorFlow Lite abandoned**	Neither LSTM nor GRU converted cleanly. Moved to ONNX, measured lossless, and kept it.
Aug 2025	**Working product**	React web and Kotlin Android clients, confidence bands, anomaly detection.
Sep 2025	**Both doors close**	Patent rejected. Google Play blocked on medical-software requirements despite a completed submission. Released as open source and set down.
Aug 2026	**Evaluated properly**	Found the flatlining, the aggregation bug, and a scoring bug that had silently disabled two thirds of the indicator system in production.

Repositories

`Atakan-Kaya35/BSP`
Training and inference containers, evaluation, model bundle

`Atakan-Kaya35/blood-sugar-predictor-web`
React and Plotly dashboard

`Atakan-Kaya35/blood-sugar-predictor-android`
Kotlin and Jetpack Compose client

`Atakan-Kaya35/bsp-privacy-policy`

Published for the Google Play submission

`Atakan-Kaya35/BSP-app-process`

The earlier Flutter attempt, April 2024

Citation

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Not a medical device. BSP is a research and educational system. It is not validated for clinical use, must not inform treatment decisions, and has no regulatory clearance in any jurisdiction. If you have diabetes, use the device your clinician gave you.

Evaluation reproducible from the published repository. Source data is one subject's own continuous glucose record, published with the code.